1. Measurement System

Truthworthy chain

- accuracy, power/portablity, calibration burden, safety/usability

Measurand

- what exactly are we measuring, the physical quantity, property, or condition the system measures.
- **Common Catagories:** biopotential, pressure, flow, imaging/dimensions, displacement, impedance, temperature, chemical concentration

Measurand chain

- measurand → transduction → conditioning → integration → performance → safety

Convert chain

- **Transductor:** Converts one form of energy to another.
- **Sensor:** A transducer that converts a physical measurand to an electrical output. (Ideally responds to the measurand)
- **Convert Chain:** Measurand → Primary sensing element → variable-conversion element → electrical signal.

Conditioning

- **Raw signal:** small + noisy.
- **Amplify:** make it usable.
- **Filter:** reduce unwanted content.
- **Match/isolate:** interface safely.
- **Digitize:** prepare for processing.
- **Display:** waveform or number.

The same chain applies to ECG, pulse oximetry, temperature, pressure, and other instruments.

Generalized System

- **Measurand:** calibration
- **Sensor/Transducer:** feedback/ control
- **Signal Conditioning:** storage
- **Output/Display:** transmission

Why biomedical measurement is hard

- Signals are often μV–mV, slow, and close to DC.
- The measurand may be inside the body or hard to isolate; patient, posture, motion, and time create biological variability.
- **Interfering input:** unwanted input, e.g. 50/60 Hz mains field.
- **Modifying input:** changes the measurement system, e.g. cable orientation or electrode–skin impedance changing mains coupling.

Typical measurands

Measurand	Typical range	Frequency range
ECG	0.5–4 mV	0.01–250 Hz
EEG	5–300 μV	dc–150 Hz
Arterial BP (direct)	10–400 mmHg	dc–50 Hz
Respiratory rate	2–50 breaths/min	0.1–10 Hz
Body temperature	32–40 °C	dc–0.1 Hz

Eg: Pulse oximeter chain

- Measurand: arterial oxygen saturation (SpO_2), inferred at the fingertip.
- Red + infrared light through tissue → photodetector current → amplify/filter/isolate the pulsatile component → compute SpO_2 from a two-wavelength ratio.
- Performance limits: motion artifact and low perfusion.

2. Static Characteristics

Accuracy and precision

- **Accuracy:** closeness to the true value.
- ±% of **full scale**: fixed absolute error across the range.
  BP range 0–300 mmHg, ±1% full scale ⇒ ±3 mmHg.
- ±% of **reading**: error grows with the reading.
  Pulse oximeter ±2% reading: at 100%, ±2%; at 50%, ±1%.
- **Precision:** repeatability; high precision does not imply high accuracy.
- **Repeatability:** same conditions, same short time window.
- **Reproducibility:** conditions vary (operator, time, instrument).

Range, span, sensitivity

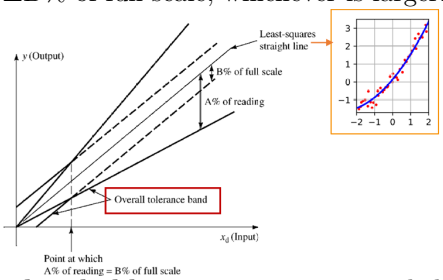
- **Tolerance:** maximum deviation of a manufactured component from a standard value.
- **Range/span:** the minimum-to-maximum designed input values.
- **Sensitivity:** slope of the calibration line:

$$S = \frac{\Delta\text{output}}{\Delta\text{input}}$$

Example: $\frac{\Delta R}{\Delta T} = \frac{7\ \Omega}{30°\text{C}} = 0.233\ \Omega/°\text{C}$.

Linearity, threshold, resolution

- **Linear:** if y_1 responds to x_1 and y_2 to x_2, then $y_1 + y_2$ responds to $x_1 + x_2$, and Ky_1 responds to Kx_1.
- **Independent nonlinearity:** maximum deviation from the least-squares line, reported as $\pm A\%$ of reading or $\pm B\%$ of full scale, whichever is larger.



- **Threshold:** minimum input needed to register any output.
- **Resolution:** smallest detectable change once the instrument is moving.

Drift, hysteresis, dead space

- **Zero drift** Δb: every output changes by the same absolute amount; slope is unchanged.
- **Sensitivity drift** Δm: slope changes; error is proportional to input magnitude.
- **Hysteresis:** loading and unloading curves do not coincide. Maximum input/output hysteresis is normally expressed as a percentage of full-scale input/output.
- **Dead space:** range of input values over which output does not change.

3. Error, Statistics, Calibration

Systematic versus random error

- **Systematic:** consistently on one side of truth; caused by loading, environmental/modifying inputs, wear, or uncalibrated instruments; calibration or correction can fix it; averaging does not.
- **Random:** scatters unpredictably about truth; averaging reduces, but never eliminates, it.
- Mean: arithmetic average. Median: middle value after sorting. Standard deviation: spread of values about the mean.

$$\bar{x} = \frac{1}{n}\sum_i x_i, \qquad \sigma = \sqrt{\frac{\sum_i (x_i - \bar{x})^2}{n-1}}$$

- Gaussian distribution probability density function

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{(x-\bar{x})^2}{2\sigma^2}\right]$$

- Gaussian rule: 68% within $\pm\sigma$; 95.4% within $\pm 2\sigma$; 99.7% within $\pm 3\sigma$.

Loading effect

- A voltmeter loads a source with finite Thévenin resistance.
- $R_{\text{source}} = 500\ \Omega$, $R_{\text{meter}} = 9500\ \Omega$:

$$\text{loading error} = \frac{R_{\text{source}}}{R_{\text{source}} + R_{\text{meter}}} = 5\%$$

Uncertainty

- Independent random uncertainties combine by quadrature addition.

$$e = \sqrt{x^2 + y^2},$$
$$e_{\text{fractional}} = \sqrt{e_1^2 + e_2^2 + \cdots}$$

- e.g. Sensor ±2% and amplifier ±1.5%: $\sqrt{0.02^2 + 0.015^2} = \pm 2.5\%$.

Calibration

- Use a known input and compare the instrument output with the reference.
- bias exists → compare to reference → calibrate/correct → reference itself traceable
- Drift is exactly why calibration has to be repeated, not done once.

4. Dynamic Characteristics

Common systems

- **Zero-order:** potentiometer, ruler
- **First-order:** thermometer, thermocouple
- **Second-order:** catherter-transducer system, acclelerometer

First-order response

- First-order system canonical form: $\tau\frac{dy(t)}{dt} + y(t) = kx(t)$, which transfer function $\frac{Y(j\omega)}{X(j\omega)} = \frac{k}{1+j\omega\tau}$, with magnitude $|H|$ and phase ϕ.

$$|H| = \frac{k}{\sqrt{1+(\omega\tau)^2}}, \quad \phi = -\arctan(\omega\tau)$$

- while the input is a step response $x(t) = m \cdot u(t)$, we have the output which is $y(t) = mk(1 - e^{-t/\tau})$.
- for $t = n\tau$, the output has:0.632(n=1), 0.865(n=2), 0.950(n=3), 0.982(n=4), 0.993(n=5), 0.998(n=6), 0.9997(n=7).
- τ is the time constant. Corner frequency: $f_c = \frac{1}{2\pi\tau}$.
- For $\tau = 20$ ms and amplitude error under 5%: $\omega \approx 16.4$ rad/s, $f \approx 2.6$ Hz, $\phi \approx -18.2°$.
- A first-order system output can never be bigger than its input.

Second-order response

- Second-order system canonical form:

$$\frac{d^2y(t)}{dt^2} + 2\zeta\omega_n\frac{dy(t)}{dt} + \omega_n^2 y(t) = k\omega_n^2 x(t),$$

- which transfer function:

$$\frac{Y(j\omega)}{X(j\omega)} = \frac{k\omega_n^2}{-\omega^2 + j2\zeta\omega_n\omega + \omega_n^2}$$

- with parameters following:

$$\Lambda = \ln\frac{y_1}{y_2}, \quad \zeta = \frac{\Lambda}{\sqrt{4\pi^2+\Lambda^2}},$$
$$\omega_d = \frac{2\pi}{T_d}, \quad \omega_n = \frac{\omega_d}{\sqrt{1-\zeta^2}}$$

- ω_n: undamped natural frequency; ω_d: damped frequency observed in a ringing trace.
- $\zeta = 1$: critically damped; $\zeta < 1$: underdamped. Underdamped responses exaggerate peaks and ring; overdamped responses smear the waveform and understate it.
- Worked values $y_1 = 8$, $y_2 = 3$, $T_d = 0.025$ s: $\Lambda = 0.9808$, $\zeta \approx 0.154$, $\omega_d \approx 251$ rad/s, $\omega_n \approx 254$ rad/s (≈ 40 Hz).

5. Physical Transducers

Transducer

- **Definition:** a kind of component that converts one form of energy to another.
- physical input → property changes → electrical signal → excitation/output
- e.g. potntiometer, move the effective length, change the output resistance.

Resistive: strain gauge

- Stretching a wire makes it longer and thinner; its resistance rises.

$$G = \frac{\Delta R/R}{\Delta L/L} \quad \text{(gauge factor)}, \quad \epsilon = \frac{\Delta L}{L}$$

- Force/pressure → small fractional ΔR; small signal needs bridge read-out. Main limitation: temperature sensitivity.

Inductive and LVDT

- Move a core inside a coil, the inductance changes, so we have:

$$L = n^2 G\mu, (unit: Henry, H)$$

- n: turns; G: geometric form factor; μ: magnetic permeability.
- LVDT uses mutual-inductance behavior. At null, secondary signals cancel; off null, imbalance gives displacement magnitude and direction. Frictionless, high resolution, robust; needs AC drive and is sensitive to stray magnetic materials.

Capacitive

$$C = \frac{\varepsilon_0\varepsilon_r A}{x}, unit: Farad, F$$

- Vary gap x, overlap area A, or dielectric constant ε_r. Non-contact, tiny displacement, low power; limited by stray capacitance and cable effects. Read out with high input impedance.

Wheatstone bridge

$$\frac{R_1}{R_2} = \frac{R_4}{R_3} \Rightarrow U_{\text{out}} = 0(atbalance),$$

- Sensor unbalances one or more arms; quarter/half/full bridge has 1/2/4 active arms.
- Approximate output:
  quarter-bridge: $\frac{1}{4}(\Delta R/R_0)v_i$;
  half-bridge: $\frac{1}{2}(\Delta R/R_0)v_i$;
  full-bridge: $(\Delta R/R_0)v_i$.
  e.g. $R_0 = 350\ \Omega$, $v_i = 5$ V, $G = 2$, $\epsilon = 10^{-3}$: $\Delta R = 0.70\ \Omega$; outputs are 2.50, 5.00, and 10.0 mV.

Piezoelectric, thermal, optical

- Piezoelectric: generate charge by squeeze crystal:

$$q = kf$$

  fast, self-generating, wide dynamic range; high output impedance requires a charge amplifier; charge leaks, so it reports changing load, not a static load.
- **Thermistor:** resistance changes with temperature; sensitive and low cost, but nonlinear.
  Positive Temperature Coefficient (PTC) / Negative Temperature Coefficient (NTC)
- **Thermocouple:** wide temperature range; detect small signals; needs a cold-junction as reference.
- **Optical:** no electrical patient connection; ambient light, motion and calibration are limitations. Used in pulse oximetry.

Eg: Stimulator/pacemaker working principle

- **Sensor:** physical input → electrical signal
- **Actuator:** electrical command → physical action
- **Stimulator / pacemaker:** supply→ oscillator → pulse output (3 stage of pulse generator) → leads → electrodes → tissue

Eg: Actuators move air or liquid

- **Ventilator:** controller or assister; Positive or negative pressure; Time-, volume- or pressure-cycled.
- **Infusion pump and valve:** pump: drives the liquid; valve: opens or closes a flow path; a valve is a simple actuator

6. Biopotentials

Origin

- The electrode is the transducer: it converts ionic current in tissue into electronic current in the lead.
- K^+ is high inside and Na^+ high outside. The Na/K pump maintains gradients; selective K^+ leak sets the resting voltage, leaving inside negative.
- Resting v_m is about -40 to -90 mV inside relative to outside; nerve and cardiac cells typically -70 to -90 mV.
- Action potential:
  Na^+ channels open → depolarization;
  K^+ flows out → repolarization
  then brief hyperpolarization.
- Equilibrium electrial potential difference could be defined by following:

$$V_K = -\frac{RT}{zF}\ln\frac{[K]_{in}}{[K]_{out}}$$

- Surface biopotential is the summed extracellular field from many active sources, not the voltage of one cell. Synchronous, similarly oriented sources reinforce; asynchronous/opposed sources partly cancel.

Fiber action potential

- **Volume conductor:** body tissue conducts current around active cells.
- **Moving charge separation:** A propagating action potential acts as a small current dipole and drives local currents outside the fiber.
- **What the electrode sees:** A weak, attenuated projection of that field, not v_m itself. Farther away

Surface recording

- **A sum of fields:** A surface recording adds the fields of many cells through conducting tissue.
- **What makes the sum large:** Synchronous, similarly oriented (in phrase) sources reinforce;
  Asynchronous or opposed ones(different timing) partly cancel.
- **Sets amplitude:** Mass of active tissue, distance to the electrode; Motor + sensory

Bio-signal Basic Info

- **Electrocardiogram, ECG:** coordinated heart-muscle activity, summed through the chest.
  P wave = atrial depolarization;
  QRS = ventricular depolarization;
  T wave = ventricular repolarization.
- **Electroencephalogram, EEG:** Excitatory Postsynaptic Potential/Inhibitory Postsynaptic Potential (EPSP/IPSP) in the cerebral cortex. Appears as microvolts due to distant sources, attenuated by skull and scalp.

awake (activity): Beta(13-30Hz)
awake (resting): Alpha(8-12Hz)
sleep (drowsiness): Theta(4-7Hz)
sleep (deepsleep): Delta(0.5-3Hz)

- **Electromyogram, EMG:** each active motor unit + the muscle fibers produces a brief extracellular motor-unit potential. As the force increases, more effort recruits more units → a dense interference pattern.
  Their extracellular potentials overlap and sum. At strong contraction, individual units merge into a dense interference pattern.
  LOW EFFORT: few units → individual potentials visible
  MORE EFFORT: more recruitment + faster firing
  STRONG CONTRACTION: waveforms overlap → interference pattern
  VERY STRONG CONTRACTION: dense interference pattern

Signal amplitude and bandwidth

Signal	Amplitude	Frequency / method
ECG	0.5–4 mV	0.01–250 Hz; skin electrodes
EEG	5–300 μV	dc–150 Hz; scalp electrodes
EMG	0.1–5 mV	dc–10 kHz; needle electrodes

- ECG: P wave = atrial depolarization; QRS = ventricular depolarization; T wave = ventricular repolarization.
- EMG: increasing force recruits more motor units and faster firing; overlapping extracellular potentials form a dense interference pattern.
- Same ECG, different purpose → different passband: monitor 0.67–40 Hz; diagnostic ≈ 0.04–150 Hz; surface EMG 15–500 Hz; needle EMG dc–10 kHz.

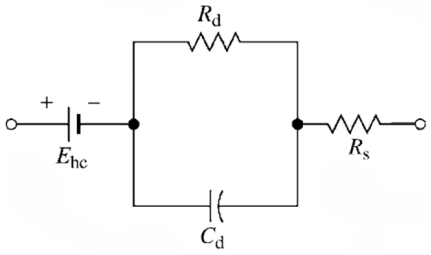
7. Electrodes

Interface chemistry

- Current is carried by electrons in metal and ions in electrolyte. Oxidation releases electrons; reduction accepts electrons.
- Charge separation creates the half-cell potential E_{hc}, set by electrode material, ion concentration, and temperature. The electrode adds interface voltage to the biological voltage.
- **Polarization:** departure of interface voltage from equilibrium when current flows; overpotential is the size of the voltage departure.
- **Non-polarizable:** no charge builds up, and charge crosses freely at steady DC (resistor-like).
- **Perfectly non-polarizable:** reactions pass charge freely and voltage stays at equilibrium. Real electrodes lie between these limits.

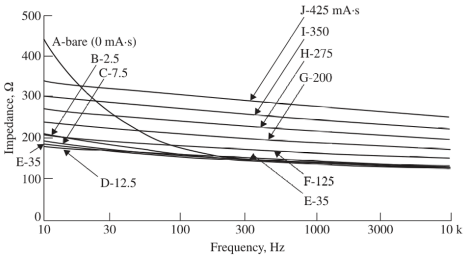
Ag/AgCl and equivalent circuit

- Ag/AgCl with chloride gel has stable interface potential, less baseline drift, and low interface impedance. It is useful because its interface stays stable, not because E_{hc} is small.
- Model: E_{hc} (battery), C_d (double-layer charge storage), R_d (leakage across double layer), R_s (interface/electrolyte resistance); $R_d \parallel C_d$.



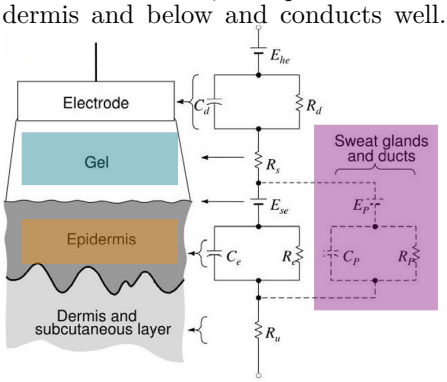
$$Z_e(f) \approx R_s + R_d \quad \text{at low } f,$$
$$Z_e(f) \to R_s \quad \text{at high } f$$

- Electrode impedance is highest at low frequency and approaches R_s at high frequency because C_d provides an easier path.
- A suitable AgCl coating lowers low-frequency impedance, but more coating is not always better.



Skin and artifacts

- Good electrode material alone cannot guarantee good contact with the body.
- Electrode provides metal contact; Gel / Sweat provides ionic contact; Epidermis is outer skin layer; its outermost part, the stratum corneum, is the main barrier; Deeper tissue is dermis and below and conducts well.



- **Already met:** E_{hc}, R_d, C_d, R_s: electrode–gel interface
- **New: epidermis** E_{se}: gel–skin potential; R_e C_e: skin resistance and capacitance.
- R_u: resistance of the deeper tissue.
- **Dashed branch:** sweat glands: an extra path (set aside).
- Skin adds impedance and can change with sweat, pressure, hair, and motion.
- Movement changes the electrode–skin interface and produces baseline wander; a filter cannot fully remove artifact that overlaps useful signal.
- Stable electrodes and good skin contact reduce motion artifact, but cannot guarantee its elimination
- Improve contact: prepare skin, secure electrode, use balanced low impedances, and reduce cable movement.

8. Amplifiers

Analog front end

- Circuits receive and prepare the electrode signal.
- **Read accurately:** Draw little current (high input impedance). Measure the voltage difference and reduce shared interference.
- **Prepare for the ADC:** Amplify within the available range (gain). Keep the frequencies needed for the ECG (bandwidth).
- **Protect the patient:** Protection limits electrical hazards. Isolation interrupts a direct conductive path to the recorder.
- The electrode offset can be much larger than the ECG signal.
  e.g. ECG input dynamic range ±5 mV; DC offset ±300 mV; input impedance at 10 Hz ≥ 2.5 MΩ (design aim ∼ 10 MΩ); system noise ≤ 30 μV peak-to-peak.

Operational amplifier

- An op amp amplifies the difference between its inputs; feedback sets a usable gain.
- Amplifies the difference between its two inputs by an enormous factor; feedback through external resistors sets a useful, predictable gain.
- **Amp working rules:**
  1. When the op-amp output is in its linear range, the two input terminals are nearly equal.
  2. No current flows into either input terminal of the op amp.
- The ECG differs between the two electrodes (differential voltage v_d); interference is nearly the same at both (common-mode voltage v_{cm}).

$$v_d = v_2 - v_1, \quad v_{\text{cm}} = \frac{v_2 + v_1}{2}$$

- **Common mode rejection ratio (CMRR):** the ability of Op amp to amplify differential signal and reject Common Mode signal.

$$\text{CMRR} = \frac{G_d}{G_c}$$

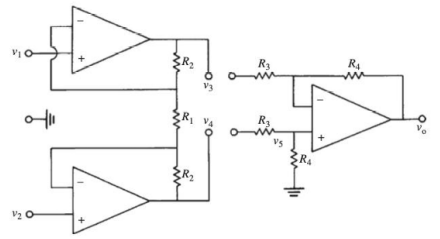
- Differential signal is wanted ECG; common-mode signal is nearly equal interference on both leads. A good biopotential amplifier has CMRR above 10,000.
- Large electrode DC offset limits gain before offset removal. For a 0–3.3 V, 12-bit ADC and 4 mV ECG, $G_{\text{total}} \leq 825$; illustrative headroom choice $G_{\text{total}} \approx 600$.
  e.g. Common-mode on the body

$V_{cm} \approx 10mV$; target $\leq 10\ /\mu V$ referred to the input
$\frac{V_{cm}}{CMRR} \leq 10\mu V \rightarrow CMRR \geq 1000$
$\rightarrow 20 \cdot log_{10}(1000) = 60$ dB.

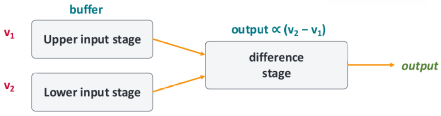
- At poor enviroment, $V_{cm} \approx 50mV$, which means CMRR ≥ 5000 or 74 dB.

Amplifier cascade

- An op-amp senses the difference, but wired simply it drains the tiny electrode signal.
- A buffer copies its input voltage while drawing almost no current, so the high-resistance electrodes are barely loaded.



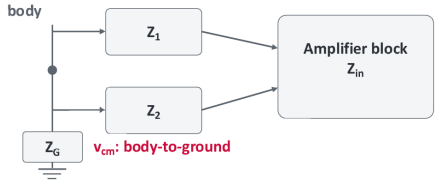
- Why not just an op amp?
  A plain one-op-amp difference circuit has low input resistance; skin electrodes have high, unequal resistance, so it drains and unbalances the signal, undoing the rejection we just built.



The instrumentation amplifier make two input buffers give very high input impedance, the difference stage rejects the shared interference, and one external resistor sets the gain all in one chip (like LT1920, AD620)

Input impedance

- Unequal electrode impedances turn common-mode voltage into a differential error.



- high amplifier CMRR is not enough, unequal electrode impedances with finite Z_{in} convert common-mode interference into a differential error.
- e.g. $V_{cm} \approx 10mV; target \leq 10\mu V$
  $Z_1 - Z_2 = 20k\Omega, Z_{in} = 5M\Omega$
  $V_d \approx V_{cm} \cdot (\frac{Z_2 - Z_1}{Z_{in}}) \approx 40\mu V > 10\mu V$
- **How to design:** use high input impedance, keep the electrodes balanced, and reduce the common-mode voltage.

9.Isolation and protection

Why needs barrier

- patient-electrodes-**BARRIER**-mainpowerinstrument-AC(220v50Hz)
- **What create risk:** electrodes make an electrical connection to the patient, so a fault in the mainspowered instrument must not open a direct current path through that connection.
- The isolation barrier passes the measurement while interrupting a direct route for a mains-side fault current

Where is barrier

- **Patient side:** protection, preamplifier, protection receives the electrode signal; the preamplifier raises the small ECG.
- **ISOLATION BARRIER:** the measurement passes; references stay separate.
- **Recorder side:** driver amplifier, ADC and microcontroller, the driver passes the signal to the ADC for digitizing.
- The recorder receives the signal across the barrier while its electrical reference stays separate.

Barrier function

- **Patient side:** instrumentation amplifier → unity-gain stage own supply and ground
- **ISOLATION BARRIER**
- **Recorder side:** separate supply and ground, mains-referenced
- What crosses the barrier: information about the measured ECG voltage.
- What stays separate: the patient-side and recorder-side supplies and grounds.
- The signal crosses; a direct fault-current path does not.

10. Filters, Noise, Interference

Filters

$$f_c = \frac{1}{2\pi RC} = \frac{1}{2\pi\tau}$$

- High-pass removes electrode DC offset and slow drift: monitor ECG 0.67 Hz; diagnostic ECG 0.04 Hz.
- Low-pass limits bandwidth and cuts high-frequency corruption/anti-alias: monitor ECG 40 Hz; diagnostic ECG 150 Hz.
- Band-pass combines high-pass and low-pass. Notch removes one frequency; 50 Hz mains notch is a last resort because 50 Hz lies inside diagnostic band.
- Higher high-pass corner rejects more motion artifact but distorts slow features; choice depends on monitor versus diagnostic purpose.

Where corruption comes from

- **Intrinsic noise:** thermal and $1/f$ noise; use a low-noise first stage and limit bandwidth.
- **External interference:** 50 Hz mains, EMI, ground loops; shield, twist, ground correctly, use driven-right-leg and high input impedance.
- **Physiological/interface artifact:** EMG, motion, offset saturation; electrode choice/placement, skin preparation, gain staging.

Diagnosis and mitigation

- **Periodic mains pickup:** periodic ripple at mains frequency → mains coupling: shield/twist leads, improve electrode contact, DRL; notch last.
- **Movement/interface artifact:** slow baseline wander with movement → secure interface, fixation/skin preparation.
- **Muscle activity or EMI:** irregular high-frequency bursts → EMG or radiated EMI: move off muscle or apply appropriate EMI filtering.
- **Clipping and saturation:** flattened peaks → offset × gain exceeds linear range: lower gain, then AC-couple.
- Electric-field coupling: shielded leads grounded at instrument, lower skin – electrode impedance.
  Common-mode on the body: high Z_{in}, balanced electrodes, driven-right-leg.
  Magnetic induction: twist wires and keep them close to body.
  Ground loop: one ground point for all instruments on the patient.
  EMI: small capacitor across the inputs; amplifiers with built-in EMI filtering

Eg: Driven-right-leg

- Sense common-mode voltage, invert and amplify it, then feed it back to the right-leg electrode; feedback drives v_{cm} down.
- It effectively grounds the patient without a direct ground connection.
- e.g. 100 kΩ right-leg electrode appears as ≈ 249 Ω; v_{cm} falls from 10 mV to ≈ 50 μV.

11. Sampling and ADC

Sampling and aliasing

$$f_s \geq 2f_{\max}$$

- Sampling measures voltage at regular instants. Below Nyquist, high frequencies fold down into false low frequencies (aliasing).
- An analog anti-aliasing low-pass filter must act **before** the ADC. For monitor ECG $f_{\max} = 40$ Hz $\Rightarrow f_s \geq 80$ Hz; diagnostic $f_{\max} = 150$ Hz $\Rightarrow f_s \geq 300$ Hz. At 400 samples/s, Nyquist is 200 Hz.

Quantization and resolution

$$q = \frac{V_{\text{ref}}}{2^{N_b}}, \qquad e_{\max} = \frac{q}{2}$$

- ADC rounds input to the nearest available level. A 12-bit, 3.3 V ADC has 4096 codes, $q = 3.3/4096 \approx 0.806$ mV, maximum error $e_{max} \approx 0.403$ mV.
- With total gain $G \approx 600$, electrode-referred step $q/G \approx 1.34$ μV. A 1 mV electrode change becomes 600 mV, or about $600/0.806 \approx 745$ ADC steps.